

UCI – Bike Sharing data

Source: [Bike Sharing - UCI Machine Learning Repository](#)

Case Study – steps

1. Download the zip file and read the ReadMe note to familiarise myself with the characteristics of the data as well as general background information
2. Open the day-based .csv in PowerBI to profile the data
3. Added columns, removed summarisation of integer fields to allow for measures calculating averages, translation to month numbers/names and transforming season from number to name
4. Started with a daily line graph for total rentals, then added totals for casual and registered; looked at seasonality, average temperature and other factors
5. Found 2 drops to investigate: early September 2011 (early hurricane season but average temperatures and other conditions still pointing to 'summer season') and end of October 2012 (Hurricane Sandy). Early September 2011 easily calls for a 10-year anniversary of 9/11 link, especially as numbers drop significantly, but a drop on 6 and 7 September points to other circumstances and a drop in temperature but more significantly a spike in humidity is the more likely cause of the drop in demand
6. Created subtitles for graphs to show thinking and explain the scatter plot for non-technical users
7. Created an executive summary and a text linking page 1 to the 2 weather-related events on page 2 for a deeper understanding of what can affect seasonality trends